AWS CLOUD • PART III

Cloud Economy & Pricing

A practical introduction to AWS pricing and cost estimation

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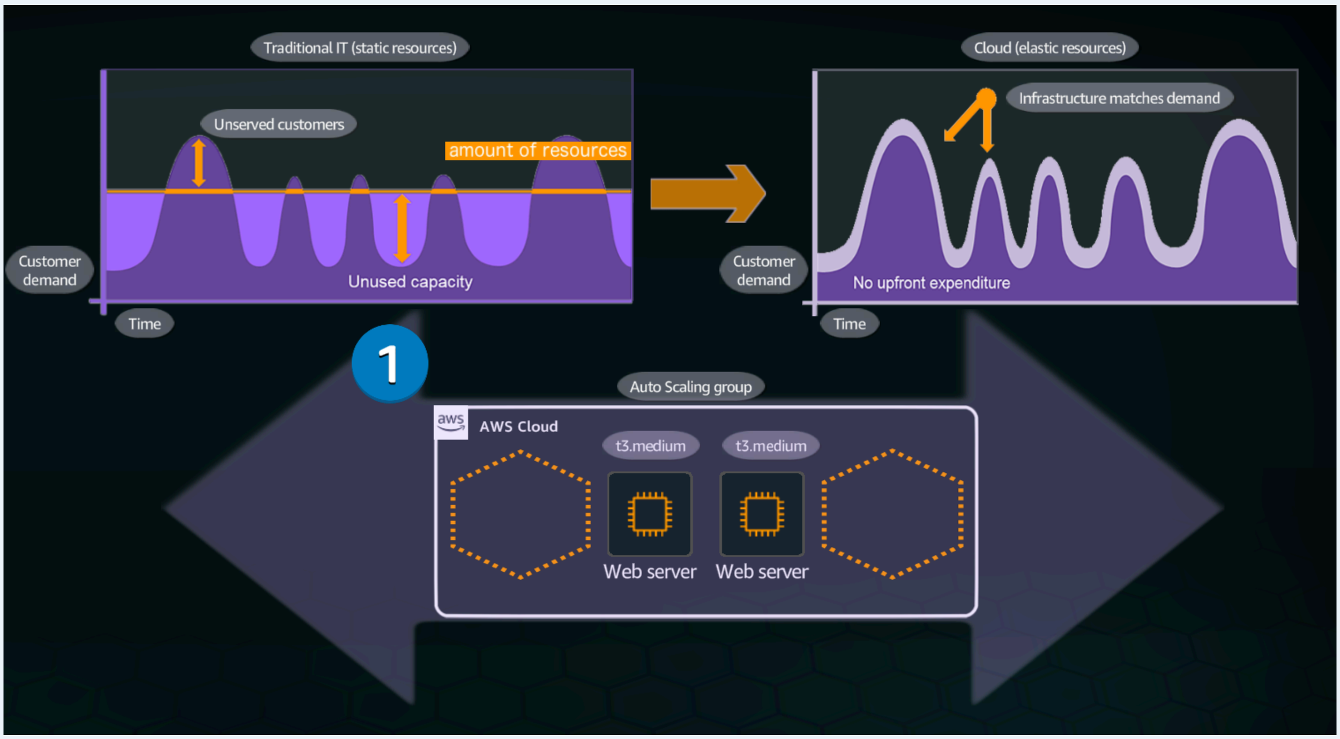
01 Cloud Economy

Cloud economics is about matching infrastructure cost to actual use. Instead of buying enough physical hardware for every possible peak, AWS lets us consume resources when needed and estimate their cost before deployment.

AWS Pricing Calculator

[1] creates estimates for AWS use cases. It can be used to model a solution, explore price points and review a detailed cost calculation before implementation.

AWS Pricing Calculator Overview



Use the calculator before deployment to explore how architectural choices affect estimated cost.

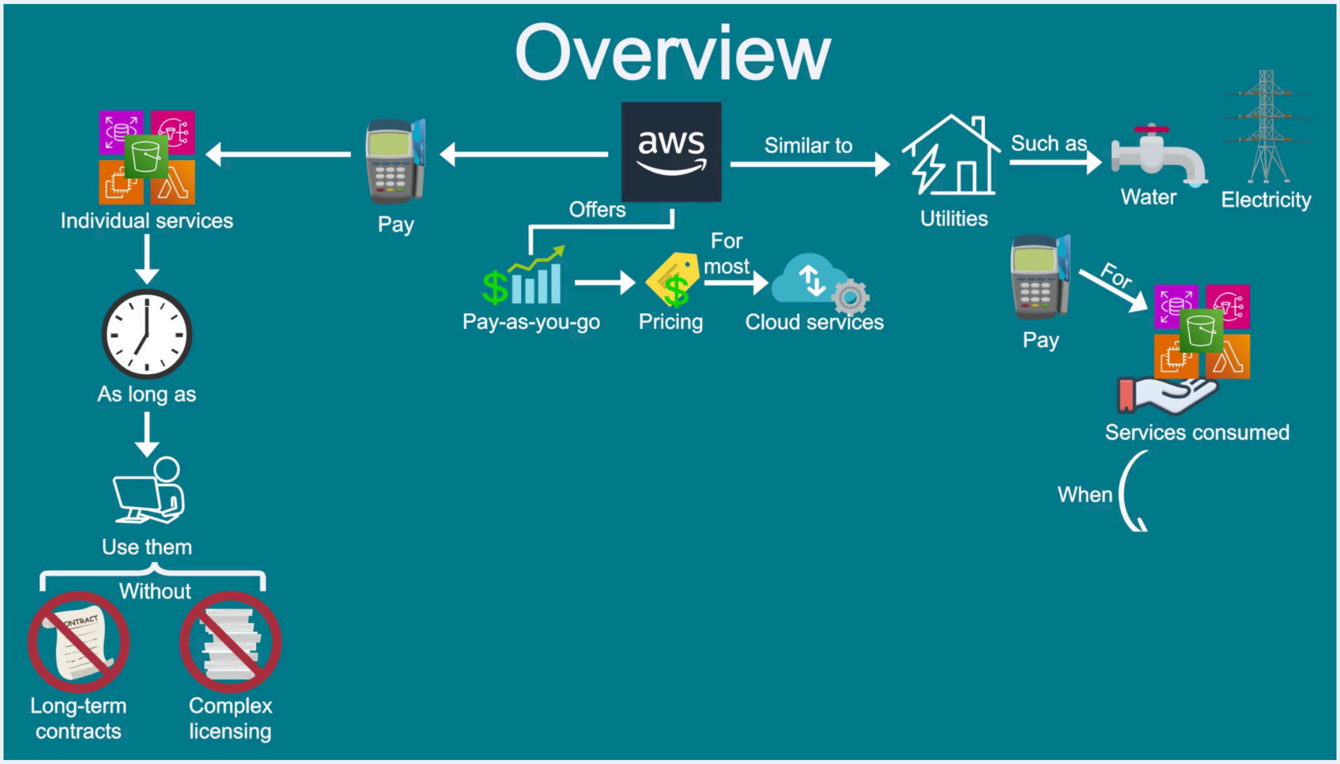
AWS - Pay as you go

AWS uses a pay-as-you-go approach for most cloud services. You pay for the individual services you need, for as long as you use them, without having to buy the underlying infrastructure.

The idea is similar to utilities such as water or electricity: usage changes, so the bill changes with it.

[1] https://calculator.aws/#/addService

Pay-as-you-go overview



Cloud spending follows resource consumption instead of requiring the full infrastructure cost up front.

Key principles

“Start early with cost optimisation”

Cloud lets us trade fixed expenses, such as data centers and physical servers, for variable expenses based on what is consumed.

Traditional	Cloud
Buy €50,000 of servers.	Use €200 of cloud resources this month.
The cost remains even when capacity sits idle.	Cost can follow actual resource usage.

Fixed vs Variable - Cost

Traditional infrastructure asks how much capacity we should buy. Cloud infrastructure asks how much capacity we need right now.

“Maximise the power of flexibility”

AWS services are priced independently and are available on demand, so resources can be selected and adjusted around the needs of the workload.

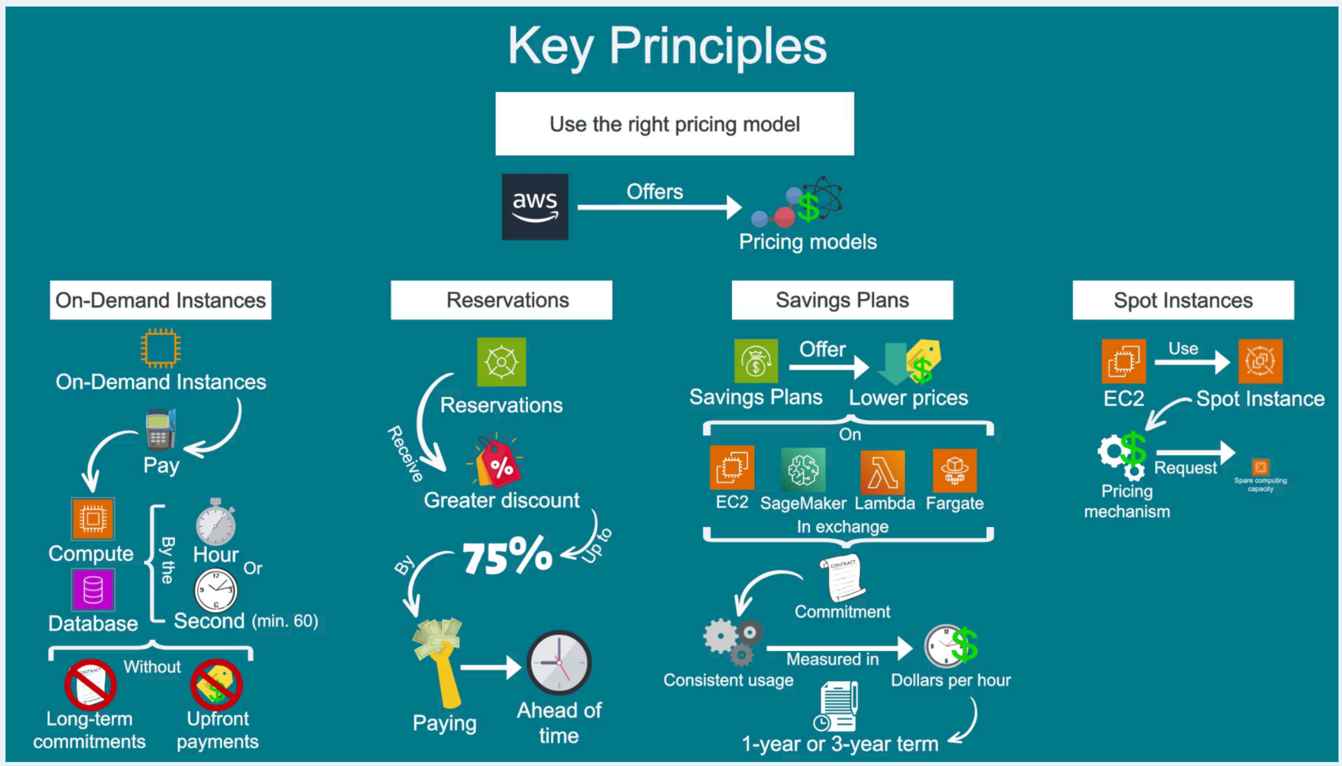
A simple example is EC2.
If an instance is not needed, stopping it stops the instance compute charge. Attached storage and some other resources can still continue to incur charges.

Cost Control - unused resources matter

Cloud makes it easier to reduce unused compute, but cost optimisation still requires monitoring what remains provisioned.

Use the right pricing model

AWS offers different pricing models depending on the service. The right choice depends on how predictable the workload is and how much flexibility you need.



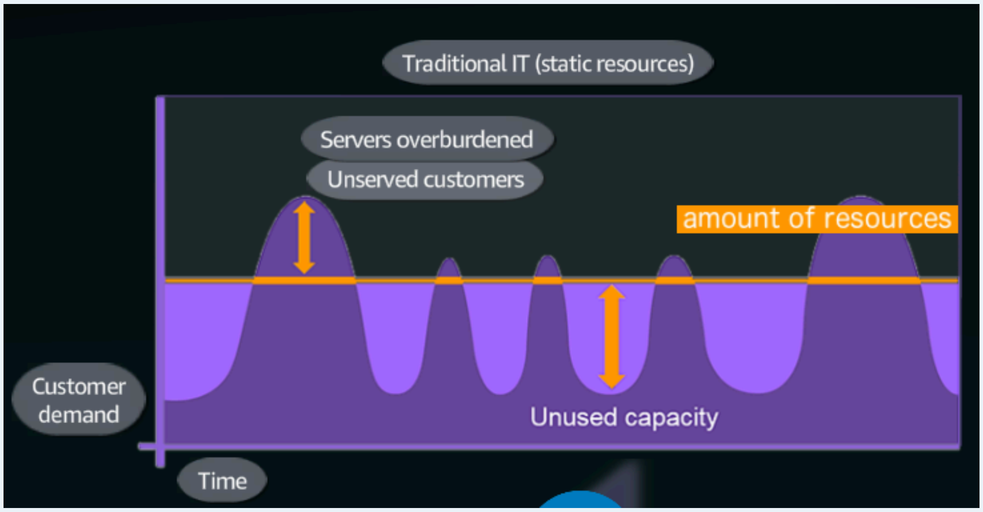
Different models exchange flexibility, commitment and spare capacity for different price levels.

Pricing model	Beginner mental model
On-Demand	Use compute without a long-term commitment and pay for the usage.
Reservations	Commit to eligible capacity or usage to receive a lower effective price.
Savings Plans	Commit to a consistent amount of eligible usage for a 1- or 3-year term.
Spot	Use spare EC2 capacity at a large discount, but the capacity can be interrupted.

Demand changes over time

Using fixed resources while demand changes creates two opposite problems. During high demand, the available capacity can become overloaded. During low demand, unused capacity creates unnecessary cost.

Demand fluctuation



Fixed capacity can be too small during peaks and too large during quiet periods.

Elasticity - Scaling capacity with demand

AWS resources can be adjusted as demand changes. During increased demand, a system can scale out horizontally by adding resources such as EC2 instances. When demand falls, active capacity can be reduced again.

[EC2, EC2] → [EC2, EC2, EC2, EC2] → [EC2, EC2]

EC2 Auto Scaling can automate this adjustment, helping infrastructure capacity follow workload demand more closely.

What to remember

- Pay-as-you-go shifts infrastructure spending toward usage-based cost.
- Flexibility lets resources change instead of remaining permanently sized for peak demand.
- Pricing models trade flexibility and commitment for different levels of savings.
- Scaling helps align capacity with demand.
- AWS Pricing Calculator estimates an architecture before it is deployed.

Use what you need, scale with demand, choose the pricing model that fits the workload, and estimate the cost before building.

AWS Pricing Calculator

The calculator does not create AWS resources or commit you to a purchase. It gives you a planning estimate so you can compare configurations, explore cost-saving opportunities and understand which parts of an architecture contribute to the expected bill.

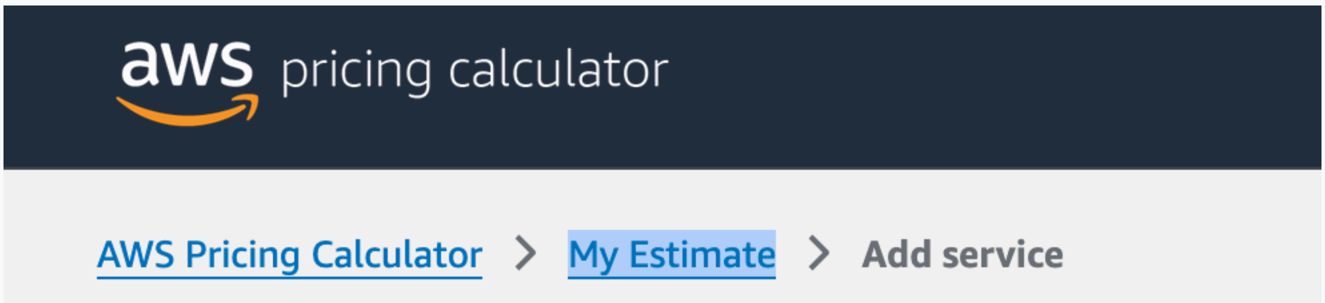
02 Practice Lab

Building an AWS cost estimate

The theory above explains how AWS pricing works. In this practical lab, we use AWS Pricing Calculator to build an EC2 web-server estimate and see how workload, resource size, pricing model, storage and data transfer affect the expected cost.

Start the estimate

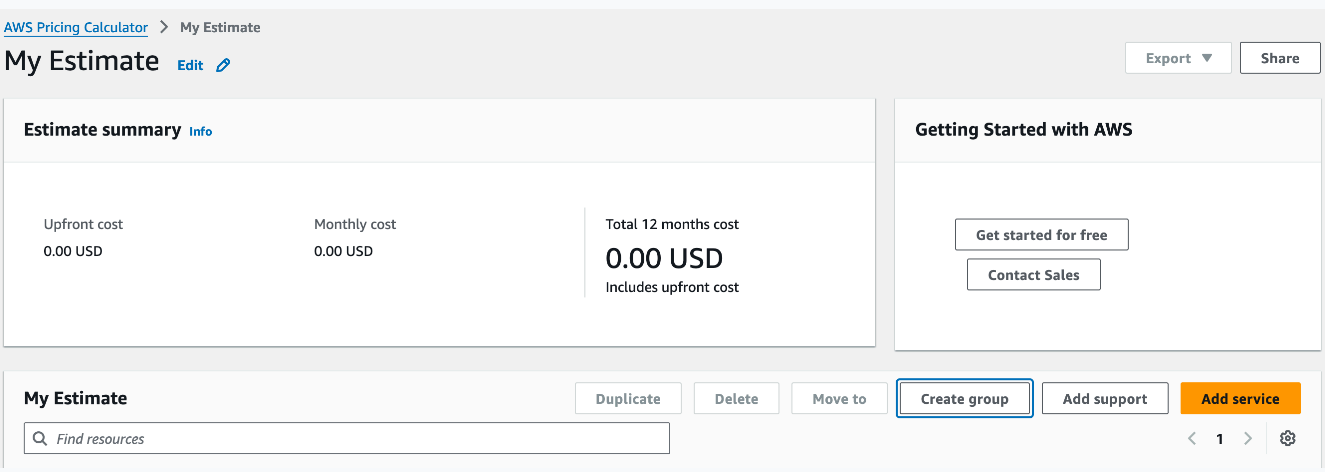
Navigate to AWS Pricing Calculator and create an estimate. The My Estimate view helps organise estimates by cost centre, product stack or solution architecture.



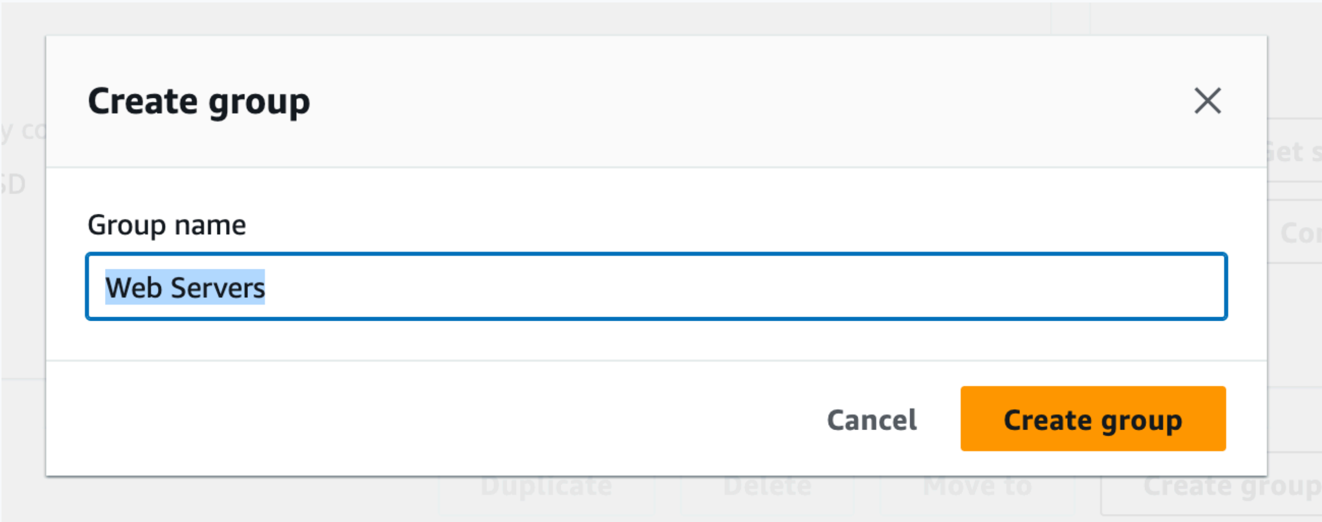
Start planning how the estimate will be organised.

Create groups

You can create different groups to compare cost estimates for different AWS setups.



Separate related services or architecture components.

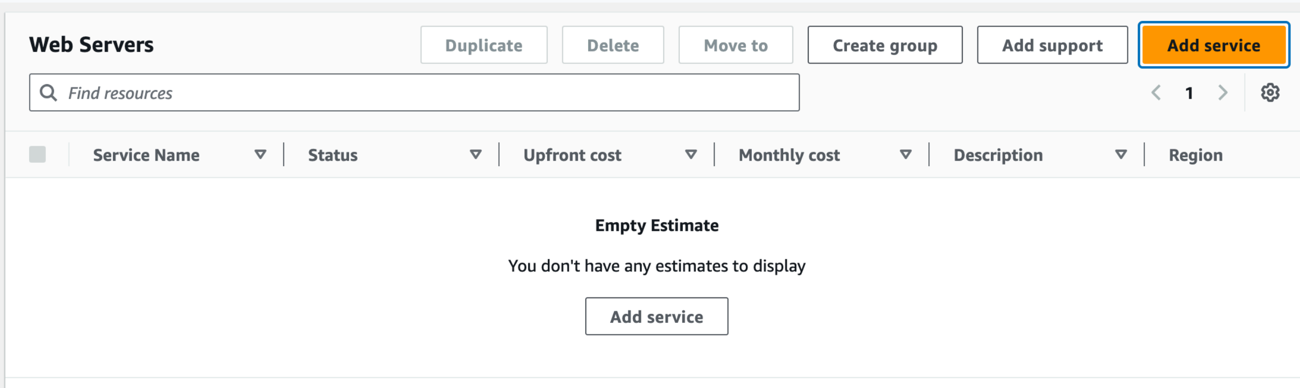


Groups make different solutions easier to compare.

Add Amazon EC2

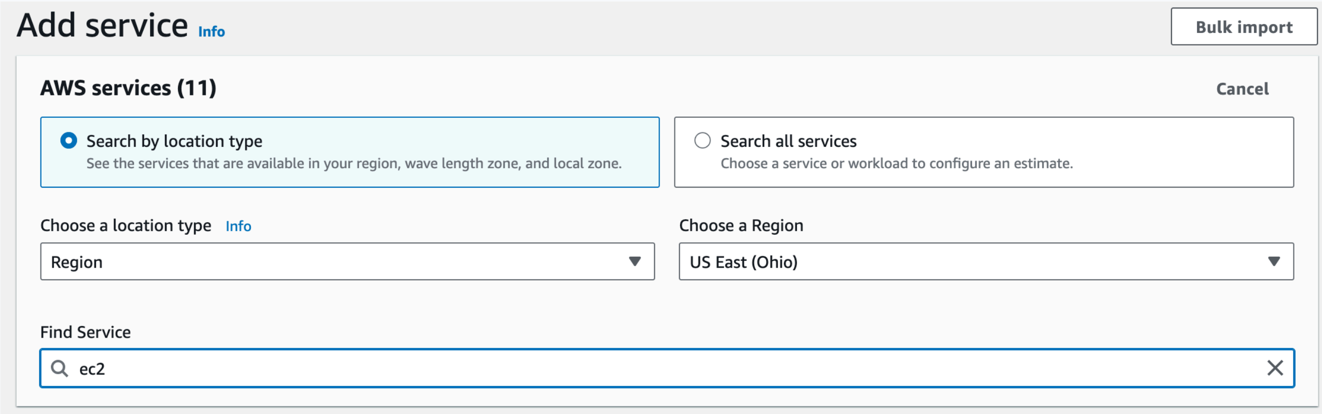
AWS Pricing Calculator provides cost estimates for more than 175 AWS services and support options. Add a new EC2 service, then choose US East (N. Virginia).

Service costs vary by AWS Region, so select the Region carefully for an accurate estimate. In this example, we estimate a web server.



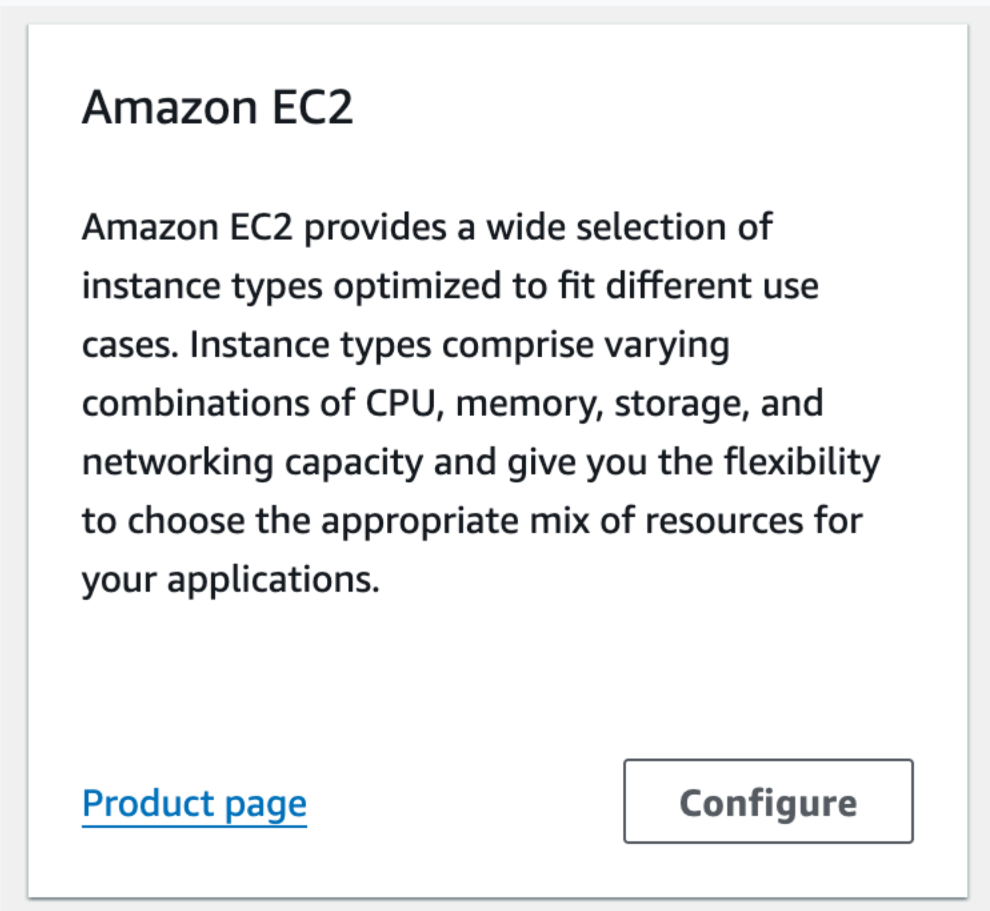
Start adding Amazon EC2 to the estimate.

Find EC2



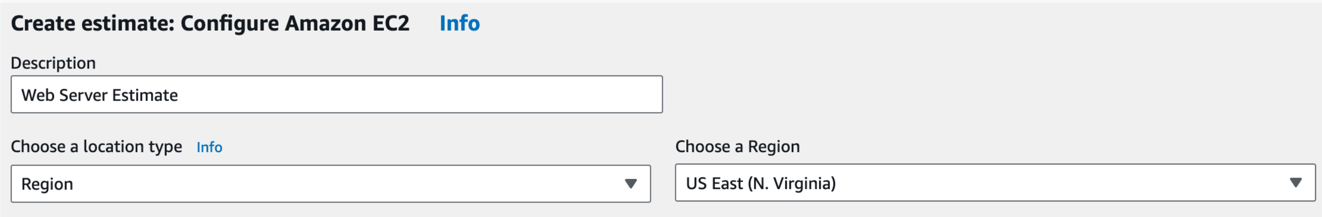
Search for Amazon EC2.

Select EC2



Open the EC2 configuration.

Choose a Region



This example uses US East (N. Virginia).

Tenancy

Our first choice is tenancy: how the underlying physical hardware is allocated.

Tenancy	Meaning
Shared / Default	Your EC2 runs on physical hardware that may also host isolated EC2 instances from other customers.
Dedicated Instance	The physical hardware is dedicated to instances from your AWS account.
Dedicated Host	You reserve an entire physical server with greater control over how instances use it.

AWS can split one powerful physical server into many isolated virtual servers. Shared tenancy lets multiple customers use the same hardware without sharing access to each other's instances. For most workloads, Shared / Default is the standard and cheaper choice.

Choose tenancy

Tenancy affects how the underlying hardware is shared and can change the price.

Workloads

A workload describes how EC2 usage changes over time. Do you need the same amount of capacity all the time, or does demand increase at certain times?

Workload	What it means	Example
Constant usage	Roughly the same capacity all month	Backend server running 24/7
Daily spike traffic	Usage increases at certain times every day	Website busy during daytime
Weekly spike traffic	Usage increases on particular days each week	Business software busiest Mon-Fri
Monthly spike traffic	Usage increases at certain times each month	Payroll system busy at month-end

Choose a workload

Model how EC2 usage changes over time.

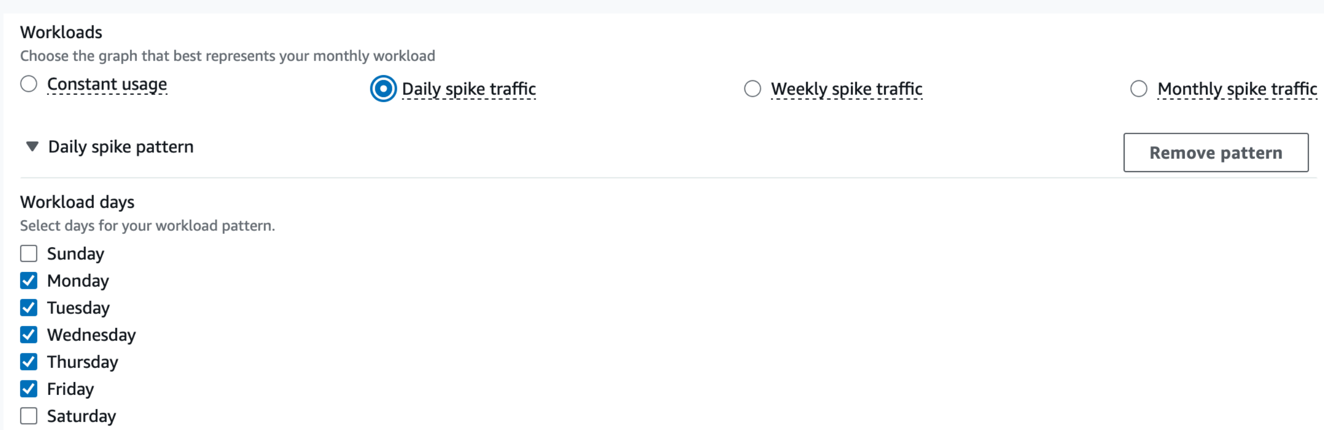
Workload example: an online store

An online store may be quiet overnight but much busier during the day and evening. That creates a daily spike because the busy hours repeat each day.

- Constant usage: relatively steady traffic throughout the month.
- Daily spike: busier during certain hours each day.
- Weekly spike: busiest on particular days, such as weekends.
- Monthly spike: activity increases around payday or month-end.

The important part is that the application type does not determine the workload pattern. How and when people use it does.

Daily spike traffic



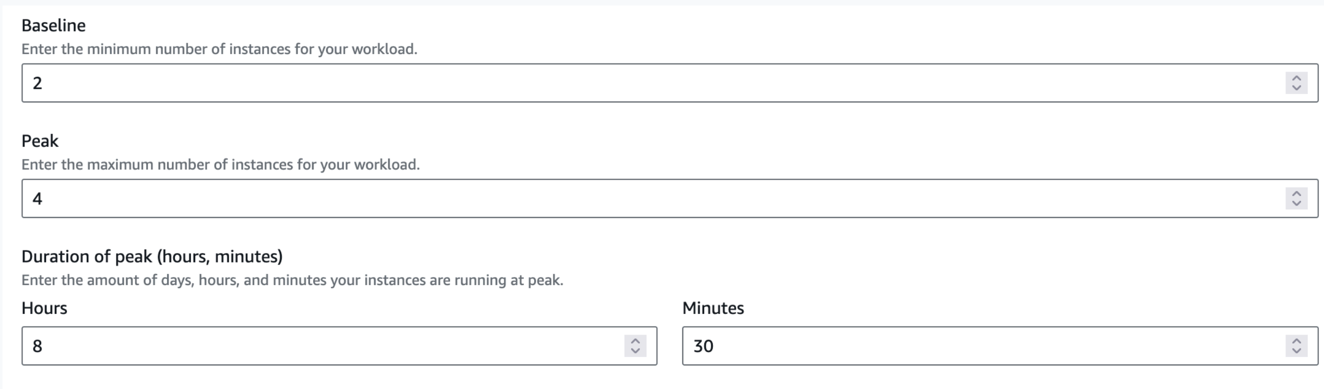
Model recurring busy hours for the online store.

Baseline and peak

- Baseline: the minimum number of EC2 instances normally needed.
- Peak: the maximum number needed when demand increases.
- Duration of peak: how long the workload stays at that maximum.

Example: the store normally runs 2 instances, scales up to 4 during busy hours, and stays at peak for 8 hours and 30 minutes.

Baseline and peak instances



Show the always-on baseline and the extra instances used during peak demand.

Estimate the resources

We also need to estimate how much compute the workload might use.

You are not expected to know the perfect values beforehand. Start with a reasonable estimate based on the workload, then monitor the real application and adjust its resources later.

Resource	What it affects	How do I know what I need?
vCPU	How much processing work the instance can handle	More users, calculations, requests or CPU-heavy work may require more vCPUs.
Memory (GiB)	How much data and app state can stay in fast working memory	Large apps, caching or many processes may require more memory.

vCPU represents virtual processing capacity, while **memory (GiB)** represents the working memory available to the instance. More vCPUs do not automatically mean a faster application; the instance family and underlying processor also affect performance.

A simple rule when starting

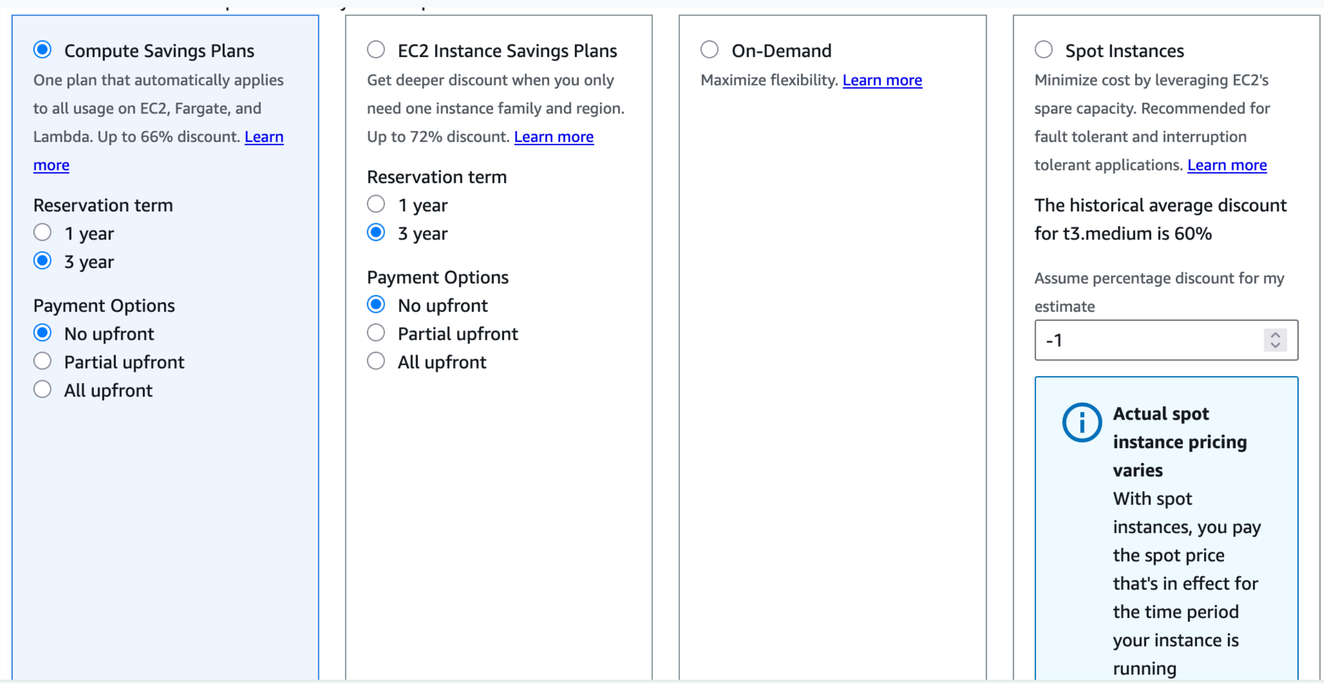
Start small and measure. CPU constantly high? Consider more vCPUs. Running out of memory? Consider more RAM. If everything runs comfortably, you probably do not need a larger instance.

You generally do not know the perfect EC2 size beforehand. Make an educated estimate, monitor the workload, and right-size it later.

How do you want to pay for EC2?

Pricing option	Simple explanation	Good for
On-Demand	Pay for what you use with no long-term commitment	Uncertain or changing workloads, learning
Compute Savings Plans	Commit to a certain amount of compute usage for 1 or 3 years in exchange for lower prices	Long-running workloads where you still want flexibility
EC2 Instance Savings Plans	Commit more closely to an EC2 instance family in a Region	Predictable EC2 workloads
Spot Instances	Use AWS spare EC2 capacity at a large discount, but AWS can interrupt it	Work that can tolerate interruptions

Choose a pricing option



Compare the EC2 pricing models available for the estimate.

The easiest way to understand them

Imagine our online store.

- On-Demand: the store is new, so we pay for EC2 as we use it without a long-term commitment.
- Compute Savings Plans: usage is predictable, so we commit to consistent compute spend for 1 or 3 years in return for a discount.
- EC2 Instance Savings Plans: we are more certain about the instance family and Region, giving up some flexibility for potentially greater savings.
- Spot Instances: use spare AWS capacity cheaply, but only for workloads that can tolerate interruption.

Useful mental model

More commitment -> less flexibility -> potentially greater savings. Spot is different: the discount comes from using spare capacity AWS can take back.

No upfront, Partial upfront and All upfront describe when the commitment is paid. Paying more upfront can result in a lower effective price.

Reserved Instances can also provide billing discounts when you commit to 1-year or 3-year terms based on selected instance specifications.

Review workload hours

You can review workload hours per day and total instance utilisation per month. In this example, instances 1-2 run 24 hours per day while instances 3-4 are used only during the daily peak.

Navigate to workload hours

▼ **Show calculations**

Utilization summary

View a complete calculation of your estimated workload hours.

On-Demand instance hours: 1946.6666

1946.6666 On-Demand instance hours x 0.0416 USD = 80.981331 USD

On-Demand instances (monthly): 80.981331 USD

Open the estimated EC2 utilisation.

Hours per day

Workload hours per day

This table calculates the total instance utilization hours per day for your estimated daily workload.

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Baseline (count)	2	2	2	2	2	2	2
Peak (count)	4	4	4	4	4	4	4
Minutes	0	0	0	0	0	0	0
Hours	8	8	8	8	8	8	8
Total duration in hours	8	8	8	8	8	8	8
Baseline+Peak	4	4	4	4	4	4	4

See when baseline and peak instances are active.

Monthly utilisation

Workload hours per month

This table calculates the total instance utilization hours per month for your estimated daily workload.

< 1 ... >

Instance c...	Sunday h...	Monday ...	Tuesday ...	Wednesd...	Thursday ...	Friday Ho...	Saturday ...	Total hours pe...	Total hours per ...
1	24	24	24	24	24	24	24	168	730
2	24	24	24	24	24	24	24	168	730
3	8	8	8	8	8	8	8	56	243.3333
4	8	8	8	8	8	8	8	56	243.3333

Compare always-on baseline instances with temporary peak instances.

Amazon Elastic Block Storage

EBS is persistent storage for EC2 instances. Think of it like a virtual SSD or hard drive attached to your EC2 server.

gp3 (General Purpose SSD) is a common EBS volume type that provides a balance between price and performance for general workloads such as web servers.

General Purpose SSD works well for basic web servers, while workloads with heavier input-output or data-transfer needs may require different storage performance.

A normal web server may occasionally read files, write logs or load application data. A busy database may perform thousands of small reads and writes, making storage performance the bottleneck even when CPU and RAM are sufficient.

Need	Why different storage might help
Normal web server	General Purpose SSD is usually enough
Busy database	May need very high IOPS
Large sequential data	May need high throughput
Cheap, infrequently accessed workloads	May prioritise cost over speed

IOPS = how many storage operations can happen per second.
Throughput = how much data can move between the EC2 instance and its EBS storage per second.

EBS performance

Storage for each EC2 instance

Choose EBS volume storage type.

General Purpose SSD (gp3) ▼

General Purpose SSD (gp3) - IOPS

gp3 supports a max of 80,000 IOPS per volume

30

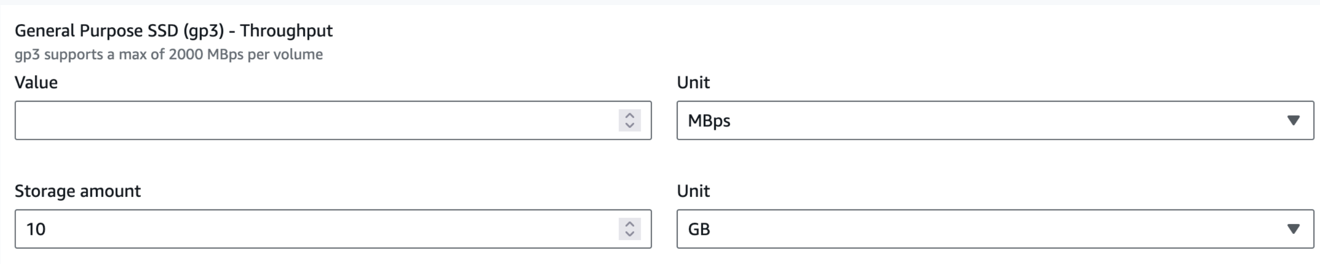
Storage performance matters differently for small operations and large data transfers.

Storage amount and throughput

When configuring an EBS volume, consider both how much data you need to store and how quickly that data needs to move. For a normal web server, reasonable defaults are usually enough to start; monitoring can tell you when to adjust later.

Setting	What it controls	How do I know what I need?
Throughput (MBps)	How quickly data moves between EC2 and EBS	Start with the gp3 default for normal applications. Increase it for sustained large reads or writes.
Storage amount (GB)	How much data the EBS volume can store	Estimate the OS, application, logs and other files, with room for growth.

EBS storage amount

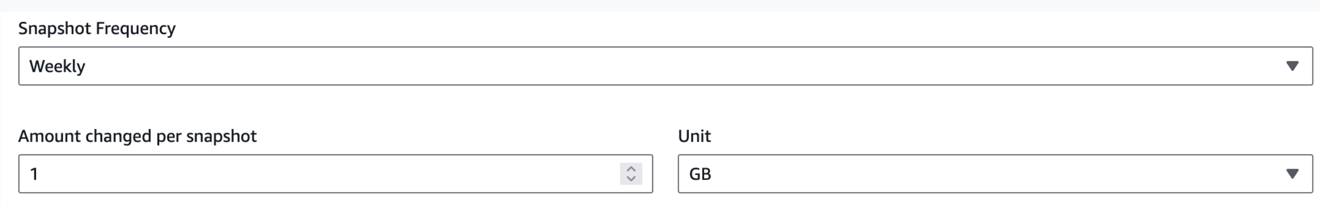


Estimate both storage capacity and performance.

Snapshot frequency

A snapshot is a point-in-time backup of an EBS volume. Snapshot frequency means how often you create that backup.

Snapshot frequency



Backup frequency is another cost input for the estimate.

Data transfer

AWS resources often need to send data to each other and to users. Some transfers are free, while others cost money depending on where the data is located and how it travels.

Simple rule: the farther data has to travel across AWS infrastructure or out to the internet, the more important it becomes to check data-transfer pricing.

Inbound & Outbound

Direction	Web app example
Inbound	A customer sends a request, form or upload to the application.
Outbound	The application sends HTML, API responses, images or files back to the customer.

Showcase

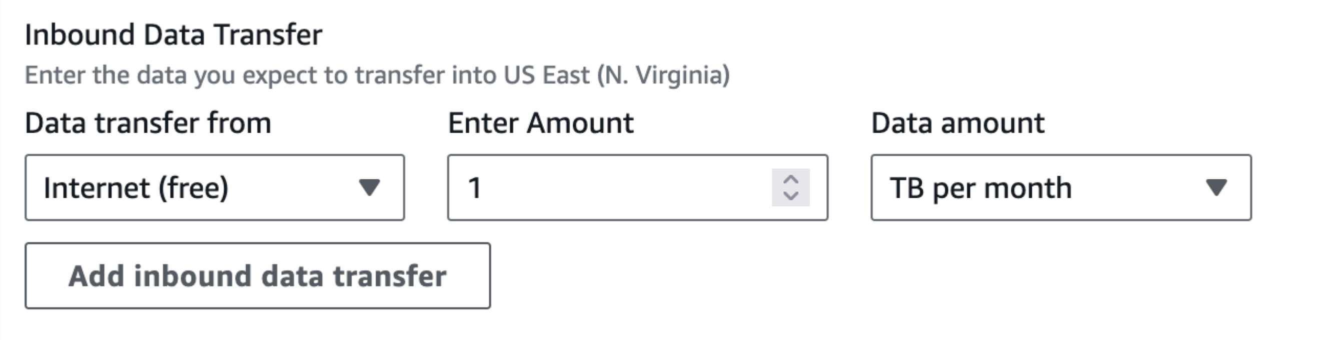
Direction	Example
Inbound	Browser requests /products
Inbound	User submits a login form
Inbound	User uploads a photo
Outbound	Server returns product data
Outbound	Server sends HTML or JSON
Outbound	User downloads a file

Inbound data

Inbound data is data travelling from the customer to your application, such as requests, forms or uploads.

Setting	What it means
Data transfer from	Where the incoming data originates, such as the internet
Enter amount	How much incoming data you expect
Data amount	The unit used to measure it, such as GB per month

Inbound data transfer



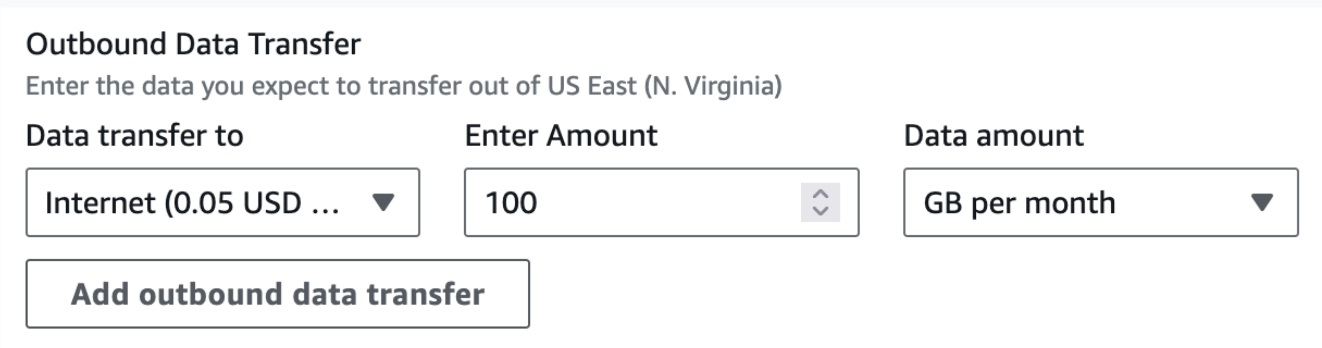
Estimate where incoming data originates and how much enters AWS.

Outbound data

Outbound data is data sent from your AWS environment to another destination. For a web app, this includes HTML, API responses, images and files returned to users.

Data transferred out to the internet can incur additional costs because AWS carries that data from its infrastructure onto external networks.

Outbound data transfer



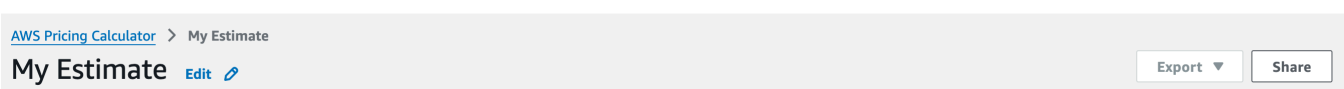
Estimate how much data the application sends from AWS to users or other destinations.

The lab's final mental model

The estimate is built from the same questions as the architecture: where does it run, how much capacity is needed, how long is it used, how is it priced, how much storage does it need, and how much data moves?

The estimate is only as accurate as these assumptions. Real usage should later be monitored and the infrastructure right-sized accordingly.

AWS Pricing Calculator turns those assumptions into an estimate before the resources are deployed.



You can finalise and share the estimates as a link.